Diego Alducin, Ph.D.

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Summary

Experienced software engineer and researcher with a Ph.D. in Physics and an M.Sc. in Computer Science, specializing in developing and deploying innovative software solutions for complex problems. Extensive expertise in data science, machine learning, physics, and cloud computing, complemented by a strong foundation in research and technical leadership. Proven ability to design scalable systems, implement advanced algorithms, and collaborate across multi-disciplinary teams to deliver impactful results. Hands-on experience creating secure high-performance applications for industries ranging from space exploration to automotive cybersecurity.

Education

University of Texas San Antonio *2024*
M.Sc. Computer Science

- **Coursework**: Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Software Engineering, Cloud Computing
- **Final Project:** "Physics-Informed Learning for Scalable Single-Qubit Control Simulations"

University of Texas San Antonio *2016*
Ph.D. Physics

- **Coursework:** Holography, Materials Science, Nanotechnology, Semiconductor Physics, Instrumentation (LabView), Diffraction Physics, Mathematical and Computational Physics.
- **Thesis Topic:** "In situ Transmission Electron Microscopy Experimentation of Nanostructured Materials."

University of Texas San Antonio *2012*
B.Sc. Physics

Experience

Senior Computer Scientist *San Antonio, TX*
Southwest Research Institute *Oct 2024 - Present*

- Developed an algorithm using object detection and advanced data analytics to efficiently identify specific events from the Juno Mission data.
- Managing a $350k project with 3 engineers to investigate the effects of Electromagnetic Interference (EMI) on electric vehicles for the National Highway Traffic Safety Administration (NHTSA).
- Continually developing business relationships with customers to provide cutting-edge solutions.
- Leading efforts to obtain funding for new projects and technologies.
- Managed a team of 4 engineers to propose a new Large Language Model (LLM) to improve the efficiency of cybersecurity testing.

Research Computer Scientist *San Antonio, TX*
Southwest Research Institute *Jan 2023 - Oct 2024*

- Implemented a secure micropatching algorithm for communications between space systems successfully deployed in the International Space Station (ISS) on an Amazon Web Services (AWS) Snowcone.
- Contributed to developing cutting-edge technology in the DARPA CRASH program. Due to NDA, details cannot be disclosed.
- Developed a web application that uses React and Redux to create cybersecurity test plans for automotive systems.
- Led the creation of an API using Python to reverse engineer CAN bus messages from vehicles directly from the OBDII port with 91% accuracy.
- Created a Neural Network to determine the future location of a vehicle based on Basic Safety messages and compared it to a physics-based predictor.
- Consulted on a project to create an anomaly detection system for space systems in their communications.

Expert Failure Analysis Engineer *San Antonio, TX*
Tower Semiconductor *May 2019 - Jan 2023*

- Established the area of Failure Analysis within the fab to streamline requests between the Engineering and Integration departments.
- Created a database web system with SQL, JavaScript, PHP and Python that ticketed requests, determined priorities, and collected data from analysis.
- Determined root failure analysis of a large family of microelectronic devices in a large-scale manufacturing environment.
- Correlated failure mechanisms of technologies between electrical measurements, process development, and physical analysis using Python and KLA Data Analysis Environment.
- Introduced document control of processes within the new area.
- Involved in quality audits for customers to determine specific production process requirements.
- Negotiated and obtained capital, $600k, for state-of-the-art instruments and upgrades.
- Responsibilities included lab and tool maintenance, management of team technicians, advise the engineering department in process improvement solutions.

Senior Process Engineer *San Antonio, TX*
Tower Semiconductor *Dec 2016 - January 2019*

- Supported production processes of 12 different technologies for large-scale production of 8-inch wafer semiconductors.
- Identified a cost-saving measure using classification algorithms which led to the introduction of a new tool to improve the lifetime of testing materials which saved $750k within a year.
- Utilized high-level data analysis such as linear and multivariate regression algorithms to determine the contribution of process changes for chemical, physical, and environmental factors in products for yield improvement and technology transfers.
- Led and managed a team of engineers that established automated control methodologies and advanced data analytics of test wafer materials and their uses, which reduced annual cost by $1.4 Million.
- Introduced process improvement projects to help maintain critical processes with Cpk above 1.67.
- Developed new processes for technology transfers with SIPOC documentation.
- Improved process control of critical tools by introducing continuous VPD ICP-MS measurements to accurately detect impurities in chemical concentrations.

Skills

Programming & Scripting: Python, C, C++, Java, JavaScript, SQL, Bash

Machine Learning & Data Science: Pandas, Numpy, PyTorch, TensorFlow, Keras, Scikit-learn, NLP, Computer Vision, Physics-Informed Neural Networks (PINNs), Object Detection, Anomaly Detection.
Web Development & Cloud Technologies: HTML5, CSS, React, Flask, Docker, AWS, GCP.
Data Engineering & Big Data: Apache Spark, Apache Airflow, Pandas, SQL.
Project Management & Collaboration: Git, GitHub, Bitbucket, Jira, Confluence, Notion.
Technical Documentation: Markdown, LaTeX, Microsoft Office (Word, Excel, PowerPoint, Access).
Languages: English (Native), Spanish (Native).

Publications and Proceedings

[1] E Gutiérrez et al. "Texture, microstructure and mechanical properties of AA7075-graphite composites produced through mechanical alloying and hot-extrusion". In: *Materials Chemistry and Physics* 296 (2023), p. 127323.

[2] Edgar Ochoa et al. "Semiconductor behavior of pentagonal silver nanowires measured under mechanical deformation". In: *Journal of Nanoparticle Research* 21 (2019), pp. 1–8.

[3] Sandra Vergara et al. "Synthesis, mass spectrometry, and atomic structural analysis of Au 2000 (SR) 290 nanoparticles". In: *The Journal of Physical Chemistry C* 122.46 (2018), pp. 26733–26738.

[4] Diego Alducin et al. "Electrical Probing of Silver Nanowires in situ Transmission Electron Microscopy". In: *Microscopy and Microanalysis* 22.S3 (2016), pp. 834–835.

[5] Diego Alducin et al. “In situ transmission electron microscopy mechanical deformation and fracture of a silver nanowire”. In: *Scripta Materialia* 113 (2016), pp. 63–67.

[6] Alina Bruma et al. “Structure determination of superatom metallic clusters using rapid scanning electron diffraction”. In: *The Journal of Physical Chemistry C* 120.3 (2016), pp. 1902–1908.

[7] Eduardo Ortega et al. “Structural damage reduction in protected gold clusters by electron diffraction methods”. In: *Advanced Structural and Chemical Imaging* 2 (2016), pp. 1–7.

[8] J Eduardo Ortega et al. “Fast scanning electron diffraction and electron holography as methods to acquire structural information on Au102 (p-MBA) 44 nanoclusters”. In: *Microscopy and Microanalysis* 22.S3 (2016), pp. 528–529.

[9] John Eder Sanchez et al. “Nano-manipulation of Ag/ZnO Nanoantennas forin-situTEM Electrical Measurements”. In: *Microscopy and Microanalysis* 22.S3 (2016), pp. 842–843.

[10] Diego Alducin, Miguel José-Yacamán, and Arturo Ponce. “On the Weak Forces on Nanoparticles”. In: *Microscopy and Microanalysis* 21.S3 (2015), pp. 955–956.

[11] Diego Alducin et al. “Kinematics of gold nanoparticles manipulation in situ transmission electron microscopy”. In: *Journal of Nanoparticle Research* 17 (2015), pp. 1–7.

[12] Brian Kiraly Joshua D. Wood Diego Alducin Benjamin D. Myers Xiaolong Liu Brandon L. Fisher Ulises Santiago Jeffrey R. Guest Miguel Jose Yacaman Arturo Ponce Artem R. Oganov Mark C. Hersam Nathan P. Guisinger Andrew J. Mannix Xiang-Feng Zhou. “Synthesis of borophenes: Anisotropic, two-dimensional boron polymorphs”. In: *Science* 350 (2015), pp. 1513–1517.

[13] Gilberto Casillas et al. “Elasticity of MoS2 sheets by mechanical deformation observed by in situ electron microscopy”. In: *The Journal of Physical Chemistry C* 119.1 (2015), pp. 710–715.

[14] Nikhita D Mansukhani et al. “High-Concentration Aqueous Dispersions of Nanoscale Two-Dimensional Materials Using Nonionic, Biocompatible Block Copolymers”. In: *Small (Weinheim an der Bergstrasse, Germany)* 12.3 (2015), p. 294.

[15] J Eduardo Ortega, Diego Alducin, and Arturo Ponce. “TEM In situ Plastic Deformation of Silver Nanowires”. In: *Microscopy and Microanalysis* 21.S3 (2015), pp. 941–942.

[16] D Bruce Buchholz et al. “The structure and properties of amorphous indium oxide”. In: *Chemistry of Materials* 26.18 (2014), pp. 5401–5411.

[17] Joohoon Kang et al. “Thickness sorting of two-dimensional transition metal dichalcogenides via copolymer-assisted density gradient ultracentrifugation”. In: *Nature communications* 5.1 (2014), p. 5478.

[18] Nikhita D Mansukhani et al. “Aqueous, high concentration dispersions of molybdenum disulfide in biocompatible block copolymers”. In: *Abstracts of Papers of the American Chemical Society*. Vol. 248. 2014.

[19] Francisco Ruiz-Zepeda et al. “Precession electron diffraction-assisted crystal phase mapping of metastable c-GaN films grown on (001) GaAs”. In: *Microscopy research and technique* 77.12 (2014), pp. 980–985.